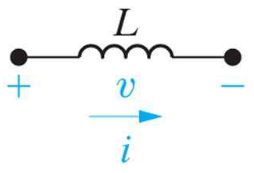
Chap.6 Inductance and Capacitance

§6.1 The Inductor



L : Henry [H]

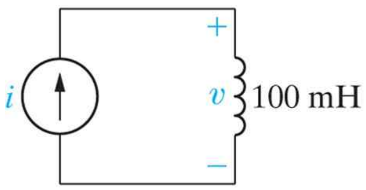
$$v(t) = L\,\frac{di(t)}{dt}$$

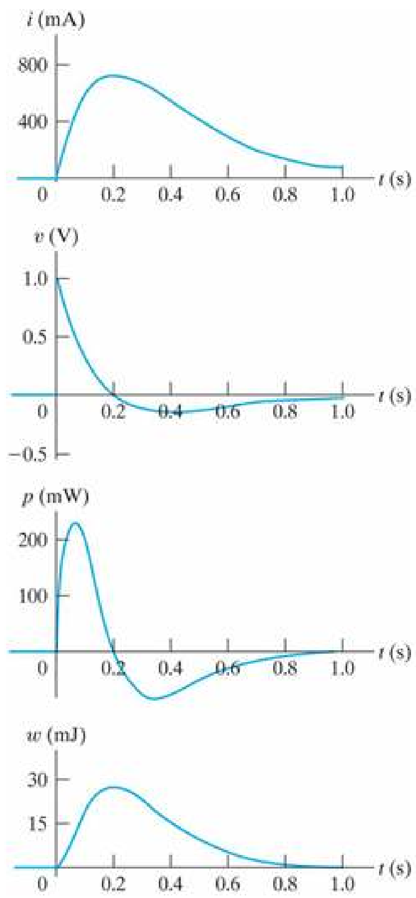
$$i(t) = \frac{1}{L}\int_{t_0}^{t} v(\tau)\,d\tau + i(t_0)$$

Power : $p(t) = v(t)\,i(t) = L\,i(t)\,\dfrac{di(t)}{dt}$

Energy : $w(t) = \dfrac{1}{2}\,L\,i^2(t)$

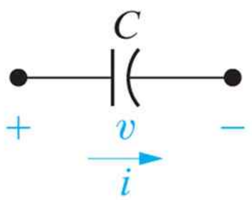
(Example)





※ The waveforms of voltage and current are different.

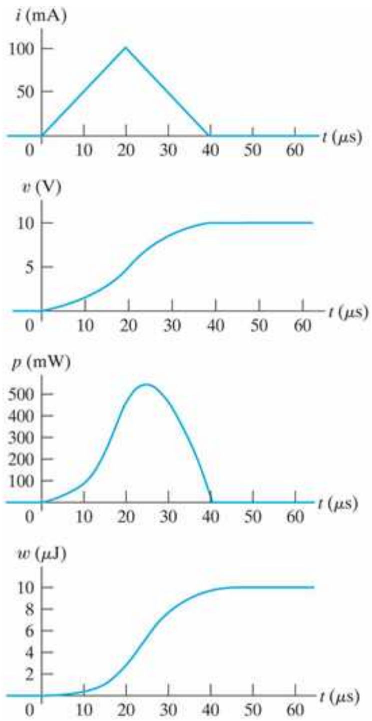
§6.2 The Capacitor

 C : Farad [F]	$i(t) = C\dfrac{dv(t)}{dt}$ $v(t) = \dfrac{1}{C}\displaystyle\int_{t_0}^{t} i(\tau)\,d\tau + v(t_0)$ Power : $p(t) = v(t)\,i(t) = C\,v(t)\,\dfrac{dv(t)}{dt}$ Energy : $w(t) = \dfrac{1}{2}\,Cv^2(t)$

(Example)

An uncharged 0.2 μF capacitor is driven by a triangular current pulse. The current pulse described by

$$i(t) = \begin{cases} 0\,, & t \le 0 \\ 5000t \ \text{A}, & 0 \le t \le 20\mu s \\ 0.2 - 5000t \ \text{A}, & 20 \le t \le 40\mu s \\ 0, & t \ge 40\mu s \end{cases}$$



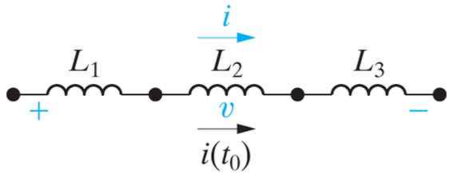
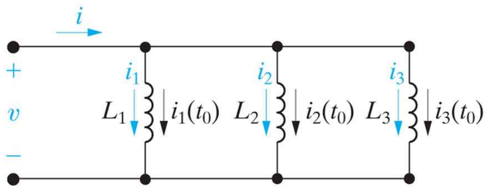
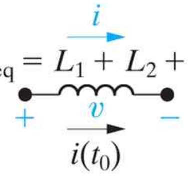
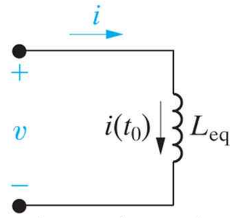
※ The waveforms of voltage and current are different.

● Continuity

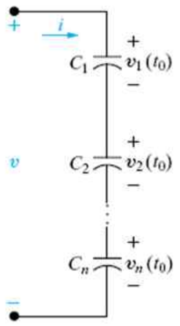
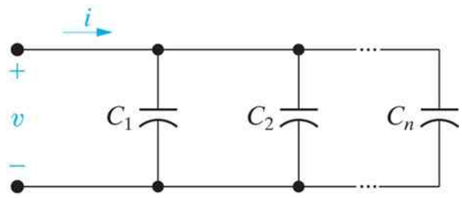
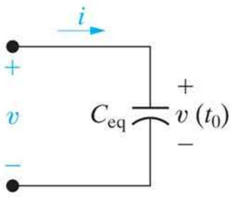
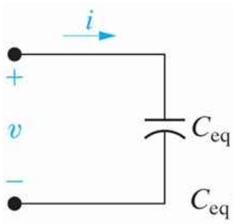
Inductor	Capacitor
$v_L(t) = L \dfrac{di_L(t)}{dt}$	$i_C(t) = C \dfrac{dv_C(t)}{dt}$
If $i_L(t)$ is jumping $\dfrac{di_L(t)}{dt} = \infty$ $\therefore$ voltage across inductor $v_L(t) = \infty$ $\Rightarrow$ impossible Therefore, the current through an inductor must be continuous at any time. $i_L(0^-) = i_L(0) = i_L(0^+)$	If $v_C(t)$ is jumping $\dfrac{dv_C(t)}{dt} = \infty$ $\therefore$ current through capacitor $i_C(t) = \infty$ $\Rightarrow$ impossible Therefore, the voltage across a capacitor must be continuous at any time. $v_C(0^-) = v_C(0) = v_C(0^+)$
Inductor; $i_L(t)$: always continuous $v_L(t)$: may or may not	Capacitor; $v_C(t)$: always continuous $i_C(t)$: may or may not

§6.3 Series-Parallel Combinations of Inductance and Capacitance

- Series-Parallel Combinations of Inductance

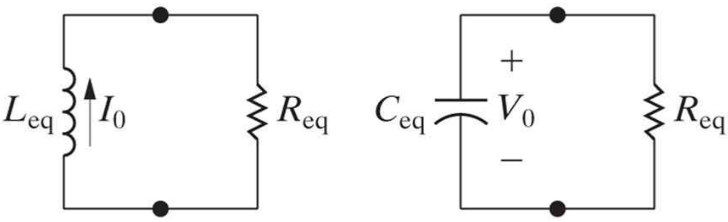
series combination	parallel combination
	
$L_{eq} = L_1 + L_2 + L_3$ 	 $\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3}$ $i(t_0) = i_1(t_0) + i_2(t_0) + i_3(t_0)$

- Series-Parallel Combinations of Capacitance

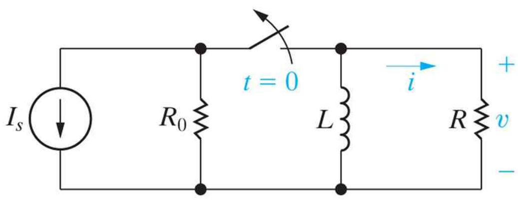
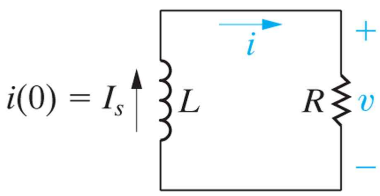
series combination	parallel combination
	
 $\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_n}$ $v(t_0) = v_1(t_0) + v_2(t_0) + \cdots + v_n(t_0)$	 $C_{eq} = C_1 + C_2 + \cdots + C_n$

Chap.7 Response of First-Order RL and RC Circuits

- the two forms of the circuits for natural response



§7.1 The natural Response of an RL Circuit

<table>
<tr><td>an RL circuit with a switch</td><td>the circuit for $t \geq 0$</td></tr>
<tr><td>
</td><td>
</td></tr>
<tr><td colspan="2">KVL (differential equation) [※ homogeneous differential eq.]

$$L\frac{di(t)}{dt} + Ri(t) = 0 \quad (\text{ initial condition: } i(0) = I_s \equiv I_0 \text{ })$$

solving the differential equation

$$i(t) = i(0)\, e^{-(R/L)t}, \quad t \geq 0$$

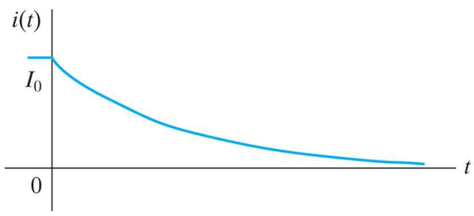
since the current in an inductor must be continuous

$$i(0^-) = i(0) = i(0^+) = I_0$$

$\therefore$ natural response : $i(t) = I_0\, e^{-(R/L)t}, \quad t \geq 0$</td></tr>
</table>

- the natural response of an RL circuit

$$i(t) = \begin{cases} I_0 \,, & t < 0 \\ I_0\, e^{-(R/L)t} \,, & t \geq 0 \end{cases}$$



※ the voltage across the resistor (i.e., across the inductor)

$$v(t) = R\,i(t) = I_0\,R\,e^{-(R/L)t}\ , \quad t \geq 0^+$$

$$v(0^-) = 0$$

$$v(0^+) = I_0\,R$$

$$v(0) = ?\ \ \text{(unknown)}$$

$$\therefore \quad v(0^-) \neq v(0^+)$$

※ The voltage across the resistor is not continuous.

- the significance of the time constant

When $e^{-(R/L)t} = e^{-1} = 0.36788 \approx 0.37$,

the time t is the TIME CONSTANT, τ .

$$\therefore \quad \tau = \frac{L}{R} \quad \text{[sec]}$$

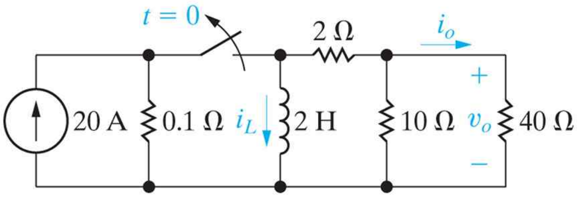
t	$e^{-t/\tau}$
τ	0.3678794
2τ	0.1353353
3τ	0.0497871
4τ	0.0183156
5τ	0.0067379

※ transient response

※ steady-state response

- Calculating the natural response of an RL circuit
  1. Find the initial current, I_0 , through the inductor.
  2. Find the time constant of the circuit, $\tau = L/R$.
  3. $i(t) = I_0\,e^{-t/\tau}\,, \quad t \geq 0$

(Example) Find $i_L(t)$ for $t \geq 0^+$.



(i) For $t < 0$, an inductor is in the steady state. (i.e., short)
20A from the current source flows through the inductor.

$$\therefore\ i_L(0^-) = 20\ \mathrm{A}$$

(ii) The current flown through the inductor must be continuous.

$$i_L(0^-) = i_L(0) = i_L(0^+) = 20\ \mathrm{A}$$

(iii) For $t \geq 0^+$, R_{eq} seen from the inductor is
(Same as the Thevenin resistance R_{th})

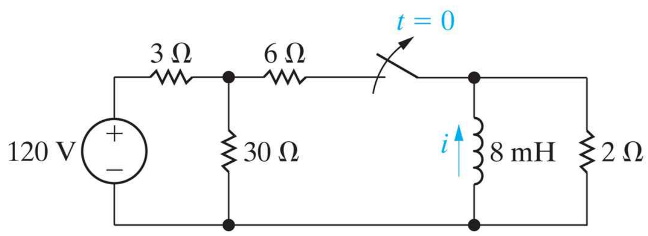
$$R_{eq} = 2 + (40 \parallel 10) = 10\ \Omega$$

(iv) time constant

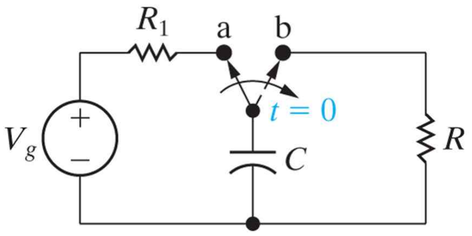
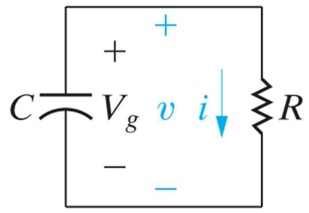
$$\tau = L / R_{eq} = (2\ \mathrm{H}) / (10\ \Omega) = 0.2\ \mathrm{sec}$$

(v) $i_L(t) = i_L(0^+)\ e^{-t/\tau} = 20\, e^{-5t}\ \mathrm{A}, \quad t \geq 0$

(Example) Find $i(t)$. [Answer: $i(t) = -12.5\, e^{-250t}\ \mathrm{A}, \quad t \geq 0$]



§7.2 The natural Response of an RC Circuit

an RC circuit with a switch	the circuit for $t \geq 0$
	

KCL [※ homogeneous differential eq.]

$$C\frac{dv(t)}{dt} + \frac{v(t)}{R} = 0 \quad \text{(initial condition: } v(0) = V_g \equiv V_0 \text{)}$$

Solving this 1st order differential equation

$$v(t) = v(0)\, e^{-t/RC}, \quad t \geq 0$$

Since the voltage across a capacitor is continuous

$$v(0^-) = v(0) = v(0^+) = V_0 = V_g$$

∴ natural response : $v(t) = V_0\, e^{-t/RC}, \quad t \geq 0$

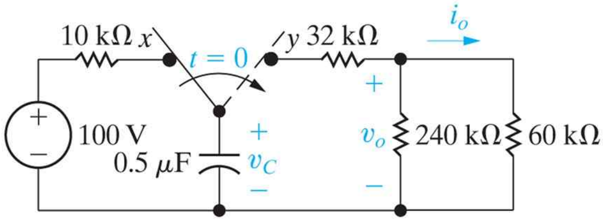
- the time constant τ for an RC circuit

  Since $v(t) = V_0\, e^{-t/RC}, \quad t \geq 0$,

  the time constant τ of RC circuit is $\tau = RC$ [sec]. (※important)

- Calculating the natural response of an RC circuit
  1. Find the initial voltage, V_0 , across the capacitor.
  2. Find the time constant of the circuit, $\tau = RC$.
  3. $v(t) = V_0\, e^{-t/\tau}, \quad t \geq 0$.

(Example) Find $v_C(t)$ for $t \geq 0^+$.



(i) For $t < 0$, the capacitor is a open circuit (steady state),
the voltage drop across the capacitor is the source voltage.

$$\therefore\ v_C(0^-) = 100\ \mathrm{V}$$

(ii) the voltage across a capacitor must be continuous, so

$$v_C(0^-) = v_C(0) = v_C(0^+) = 100\ \mathrm{V}$$

(iii) For $t \geq 0^+$, R_{eq} seen from the capacitor is
(same method to find R_{th} in Thevenin theorem)

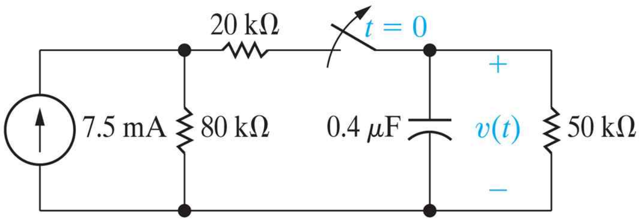
$$R_{eq} = 32\mathrm{k} + (240\mathrm{k} \parallel 60\mathrm{k}) = 80\ \mathrm{k}\Omega$$

(iv) time constant

$$\tau = R_{eq}\,C = (80\mathrm{k}\Omega)\,(0.5\mu F) = 40\ ms$$

(v) $v_C(t) = v_C(0^+)\ e^{-t/\tau} = 100\,e^{-25t}\ \mathrm{V}\,,\quad t \geq 0$

(Example) Find $v(t)$. [Answer: $v(t) = 200\,e^{-50t}\ \mathrm{V}\,,\quad t \geq 0$]



2.3 LINEAR EQUATIONS

A type of first-order differential equation that occurs frequently in applications is the linear equation. Recall from Section 1.1 that a **linear first-order equation** is an equation that can be expressed in the form

$$a_1(x)\frac{dy}{dx} + a_0(x)y = b(x) \ , \tag{1}$$

where $a_1(x)$, $a_0(x)$, and $b(x)$ depend only on the independent variable x, not on y.

For example, the equation

$$x^2 \sin x - (\cos x)y = (\sin x)\frac{dy}{dx}$$

is linear, because it can be rewritten in the form

$$(\sin x)\frac{dy}{dx} + (\cos x)y = x^2 \sin x \ .$$

However, the equation

$$y\frac{dy}{dx} + (\sin x)y^3 = e^x + 1$$

is not linear; it cannot be put in the form of equation (1) due to the presence of the y^3 and $y\,dy/dx$ terms.

There are two situations for which the solution of a linear differential equation is quite immediate. The first arises if the coefficient $a_0(x)$ is identically zero, for then equation (1) reduces to

$$a_1(x)\frac{dy}{dx} = b(x) \ , \tag{2}$$

which is equivalent to

$$y(x) = \int \frac{b(x)}{a_1(x)}dx + C$$

$[$as long as $a_1(x)$ is not zero$]$.

The second is less trivial. Note that if $a_0(x)$ happens to equal the derivative of $a_1(x)$—that is, $a_0(x) = a_1'(x)$—then the two terms on the left-hand side of equation (1) simply comprise the derivative of the product $a_1(x)y$:

$$a_1(x)y' + a_0(x)y = a_1(x)y' + a_1'(x)y = \frac{d}{dx}[a_1(x)y] \ .$$

Therefore equation (1) becomes

$$\frac{d}{dx}[a_1(x)y] = b(x) \tag{3}$$

and the solution is again elementary:

$$a_1(x)y = \int b(x)dx + C \ ,$$

$$y(x) = \frac{1}{a_1(x)}\left[\int b(x)dx + C\right] \ .$$

One can seldom rewrite a linear differential equation so that it reduces to a form as simple as (2). However, the form (3) can be achieved through multiplication of the original equation (1) by a well-chosen function $\mu(x)$. Such a function $\mu(x)$ is then called an "integrating factor" for equation (1). The easiest way to see this is first to divide the original equation (1) by $a_1(x)$ and put it into **standard form**

$$\text{(4)} \qquad \frac{dy}{dx} + P(x)y = Q(x) \ ,$$

where $P(x) = a_0(x)/a_1(x)$ and $Q(x) = b(x)/a_1(x)$.

Next we wish to determine $\mu(x)$ so that the left-hand side of the multiplied equation

$$\text{(5)} \qquad \mu(x)\frac{dy}{dx} + \mu(x)P(x)y = \mu(x)Q(x)$$

is just the derivative of the product $\mu(x)y$:

$$\mu(x)\frac{dy}{dx} + \mu(x)P(x)y = \frac{d}{dx}[\mu(x)y] = \mu(x)\frac{dy}{dx} + \mu'(x)y \ .$$

Clearly, this requires that μ satisfy

$$\text{(6)} \qquad \mu' = \mu P \ .$$

To find such a function, we recognize that equation (6) is a separable differential equation, which we can write as $(1/\mu)d\mu = P(x)dx$. Integrating both sides gives

$$\text{(7)} \qquad \mu(x) = e^{\int P(x)dx} \ .$$

With this choice[†] for $\mu(x)$, equation (5) becomes

$$\frac{d}{dx}[\mu(x)y] = \mu(x)Q(x) \ ,$$

which has the solution

$$\text{(8)} \qquad y(x) = \frac{1}{\mu(x)}\left[\int \mu(x)Q(x)dx + C\right] \ .$$

Here C is an arbitrary constant, so (8) gives a one-parameter family of solutions to (4). This form is known as the **general solution** to (4).

Method for Solving Linear Equations

(a) Write the equation in the standard form

$$\frac{dy}{dx} + P(x)y = Q(x) \ .$$

(b) Calculate the integrating factor $\mu(x)$ by the formula

$$\mu(x) = \exp\left[\int P(x)dx\right] \ .$$

(c) Multiply the equation in standard form by $\mu(x)$ and, recalling that the left-hand side is just $\frac{d}{dx}[\mu(x)y]$, obtain

$$\underbrace{\mu(x)\frac{dy}{dx} + P(x)\mu(x)y}_{\frac{d}{dx}[\mu(x)y]} = \mu(x)Q(x) \ ,$$

$$\frac{d}{dx}[\mu(x)y] = \mu(x)Q(x) \ .$$

(d) Integrate the last equation and solve for y by dividing by $\mu(x)$ to obtain (8).

2.3 EXERCISES

In Problems 1–6, determine whether the given equation is separable, linear, neither, or both.

1. $\dfrac{dx}{dt} + xt = e^x$

2. $x^2\dfrac{dy}{dx} + \sin x - y = 0$

3. $3t = e^t\dfrac{dy}{dt} + y\ln t$

4. $(t^2 + 1)\dfrac{dy}{dt} = yt - y$

5. $3r = \dfrac{dr}{d\theta} - \theta^3$

6. $x\dfrac{dx}{dt} + t^2x = \sin t$

In Problems 7–16, obtain the general solution to the equation.

7. $\dfrac{dy}{dx} = \dfrac{y}{x} + 2x + 1$

8. $\dfrac{dy}{dx} - y - e^{3x} = 0$

9. $x\dfrac{dy}{dx} + 2y = x^{-3}$

10. $\dfrac{dr}{d\theta} + r\tan\theta = \sec\theta$

11. $(t + y + 1)dt - dy = 0$

12. $\dfrac{dy}{dx} = x^2e^{-4x} - 4y$

13. $y\dfrac{dx}{dy} + 2x = 5y^3$

14. $x\dfrac{dy}{dx} + 3(y + x^2) = \dfrac{\sin x}{x}$

15. $(x^2 + 1)\dfrac{dy}{dx} + xy - x = 0$

16. $(1 - x^2)\dfrac{dy}{dx} - x^2y = (1 + x)\sqrt{1 - x^2}$

In Problems 17–22, solve the initial value problem.

17. $\dfrac{dy}{dx} - \dfrac{y}{x} = xe^x\ , \qquad y(1) = e - 1$

18. $\dfrac{dy}{dx} + 4y - e^{-x} = 0\ , \qquad y(0) = \dfrac{4}{3}$

19. $t^2\dfrac{dx}{dt} + 3tx = t^4\ln t + 1, \qquad x(1) = 0$

20. $\dfrac{dy}{dx} + \dfrac{3y}{x} + 2 = 3x\ , \qquad y(1) = 1$

21. $\cos x\dfrac{dy}{dx} + y\sin x = 2x\cos^2 x\ ,$

$$y\left(\frac{\pi}{4}\right) = \frac{-15\sqrt{2}\pi^2}{32}$$

22. $\sin x\dfrac{dy}{dx} + y\cos x = x\sin x\ , \qquad y\left(\dfrac{\pi}{2}\right) = 2$

23. Radioactive Decay. In Example 2 assume that the rate at which RA_1 decays into RA_2 is $40e^{-20t}$ kg/sec and the decay constant for RA_2 is $k = 5$/sec. Find the mass $y(t)$ of RA_2 for $t \geq 0$ if initially $y(0) = 10$ kg.

24. In Example 2 the decay constant for isotope RA_1 was 10/sec, which expresses itself in the exponent of the rate term $50e^{-10t}$ kg/sec. When the decay constant for RA_2 is $k = 2$/sec, we see that in formula (14) for y the term $(185/4)e^{-2t}$ eventually dominates (has greater magnitude for t large).

27. Consider the initial value problem

$$\frac{dy}{dx} + \sqrt{1 + \sin^2 x}\, y = x\ , \qquad y(0) = 2\ .$$

(a) Using definite integration, show that the integrating factor for the differential equation can be written as

$$\mu(x) = \exp\left(\int_0^x \sqrt{1 + \sin^2 t}\, dt\right)$$

and that the solution to the initial value problem is

$$y(x) = \frac{1}{\mu(x)}\int_0^x \mu(s)\, s\, ds + \frac{2}{\mu(x)}\ .$$

(b) Obtain an approximation to the solution at $x = 1$ by using numerical integration (such as Simpson's rule, Appendix C) in a nested loop to estimate values of $\mu(x)$ and, thereby, the value of

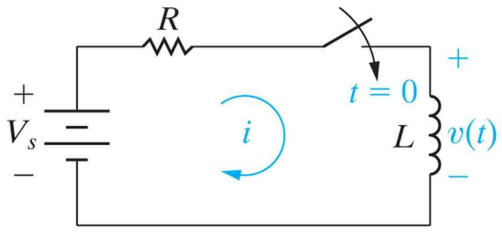
$$\int_0^1 \mu(s)\, s\, ds\ .$$

[*Hint:* First, use Simpson's rule to approximate $\mu(x)$ at $x = 0.1, 0.2, \ldots, 1$. Then use these values and apply Simpson's rule again to approximate $\int_0^1 \mu(s)\, s\, ds$.]

(c) Use Euler's method (Section 1.4) to approximate the solution at $x = 1$, with step sizes $h = 0.1$ and 0.05.

§7.3 The Step Response of RL and RC Circuits

● the step response of an RL circuit



KVL: $L\dfrac{di(t)}{dt} + Ri(t) = V_s$

[※ nonhomogeneous differential equation]

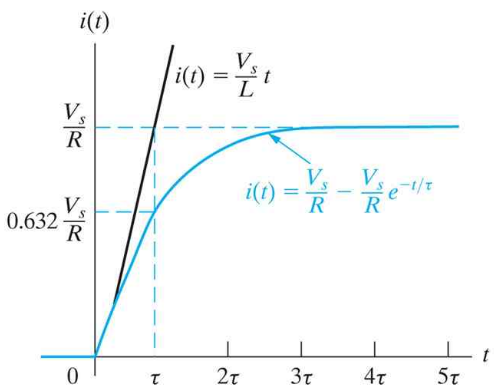
Solving the differential equation, the solution is;

$$\boxed{\begin{aligned} i(t) &= \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R} \right) e^{-(R/L)t}, \quad t \geq 0 \\ &= i(\infty) + (i(0) - i(\infty))\, e^{-(R/L)t} \end{aligned}}$$

- If the initial energy stored in the inductor is ZERO, (i.e., $I_0 = 0$)

$$i(t) = \frac{V_s}{R} - \frac{V_s}{R} e^{-(R/L)t}, \quad t \geq 0$$

$$i(\tau) = \frac{V_s}{R} - \frac{V_s}{R} e^{-1} = 0.6321 \frac{V_s}{R}$$

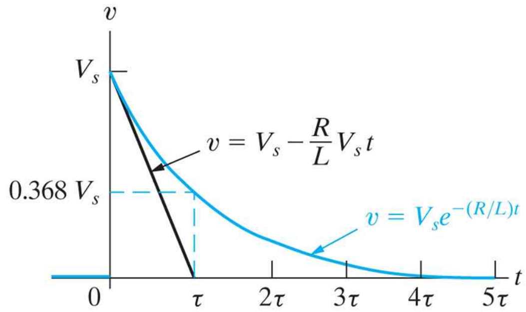


- the voltage across an inductor for $t \geq 0^+$

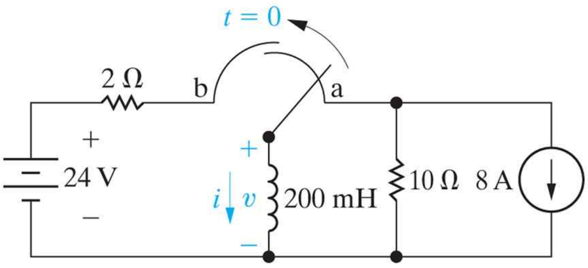
$$v(t) = L\frac{di(t)}{dt} = L\left(\frac{-R}{L}\right)\left(I_0 - \frac{V_s}{R}\right)e^{-(R/L)t}$$

$$= (V_s - I_0 R)\,e^{-(R/L)t}$$

- When $I_0 = 0$, $v(t) = V_s\, e^{-(R/L)t}$



(Example) Find $i(t)$ and $v(t)$.



(i) initial value : $i(0^-) = -8\ \mathrm{A} = i(0^+)$

(ii) final value : $i(\infty) = (24\ \mathrm{V})/(2\ \Omega) = 12\ \mathrm{A}$

(iii) time constant : $\tau = L/R = (200\ \mathrm{mH})/(2\ \Omega) = 100\ \mathrm{ms}$

(iv) current : $i(t) = 12 + (-8-12)\,e^{-t/0.1}$

$= 12 - 20\,e^{-10t}\ \mathrm{A}, \quad t \geq 0$

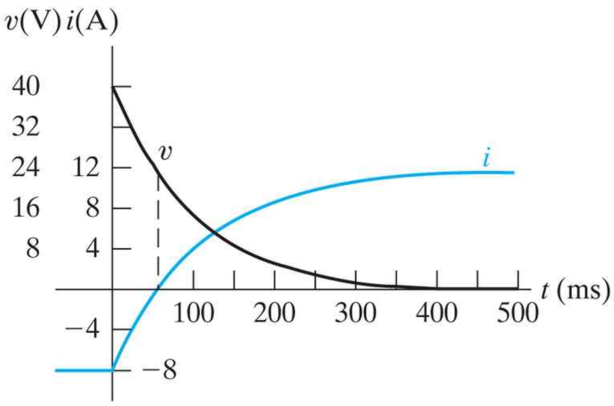
(v) voltage : $v(t) = L\dfrac{di(t)}{dt} = 0.2\,(200\,e^{-10t})$

$= 40\,e^{-10t}\ \mathrm{V}, \quad t \geq 0^+$

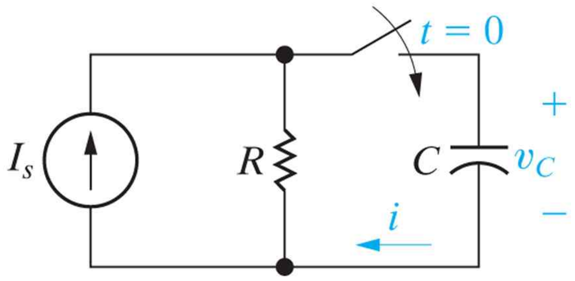
$v(0^+) = 40\ \mathrm{V}$, But $v(0^-) = 0\ \mathrm{V}$

(i.e., the voltage across an inductor is not continuous at $t = 0$)

(vi) current and voltage waveforms



● <u>the step response of an RC circuit</u>



KCL: $$C\frac{dv_C(t)}{dt}+\frac{v_C(t)}{R}=I_s$$

[※ nonhomogeneous differential equation]

Solving the differential equation, the solution is

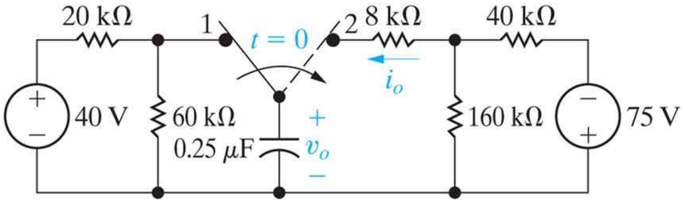
$$\boxed{\begin{aligned} v_C(t) &= I_s R + (V_0 - I_s R)\, e^{-t/RC}, \quad t \geq 0 \\ &= v_C(\infty) + (v_C(0) - v_C(\infty))\, e^{-t/RC} \end{aligned}}$$

● If the initial energy stored in a capacitor is zero (i.e., $V_0 = 0$)

$$v_C(t) = I_s R - I_s R e^{-t/RC}, \quad t \geq 0$$

$$v_C(\tau) = I_s R - I_s R\, e^{-1} = 0.6321\, I_s R$$

(Example) Find $v_0(t)$ for $t \geq 0$ and $i_0(t)$ for $t \geq 0^+$.



(i) initial condition :

$$v_0(0^-) = v_0(0) = v_0(0^+) = (40\,\mathrm{V})\left(\frac{60\,\mathrm{k}}{20\,\mathrm{k} + 60\,\mathrm{k}}\right) = 30\ \mathrm{V}$$

(ii) final value :

$$v_0(\infty) = (-75\,\mathrm{V})\left(\frac{160\,\mathrm{k}}{40\,\mathrm{k} + 160\,\mathrm{k}}\right) = -60\ \mathrm{V}$$

(iii) R_{eq} seen from the capacitor (i.e., R_{th} in Thevenin)

$$R_{eq} = 8\,\mathrm{k} + (40\,\mathrm{k} \parallel 160\,\mathrm{k}) = 40\,\mathrm{k}\Omega$$

(iv) time constant

$$\tau = R_{eq}C = (40 \times 10^3) \times (0.25 \times 10^{-6}) = 10\ \mathrm{ms}$$

(v)
$$\begin{aligned} v_0(t) &= -60 + [30 - (-60)]\,e^{-100\,t} \\ &= -60 + 90\,e^{-100t}\ \mathrm{V}, \quad t \geq 0 \end{aligned}$$

(vi)
$$i(t) = C\frac{dv_0(t)}{dt} = -2.25\,e^{-100t}\ \mathrm{mA}$$

§7.4 A General Solution for Step and Natural Responses

- general solution for natural and step responses of RL/RC circuits

$$x(t) = x_f + (x_0 - x_f)\,e^{-t/\tau}$$

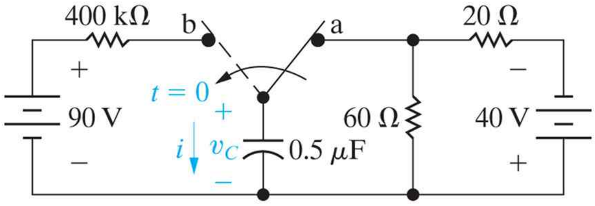
$x(t)$: the unknown variable as a function of time

x_f : the final value of the variable

x_0 : the initial value of the variable

τ : the time constant

(Example)



(i) the initial value of $v_C(t)$: $v_C(0^-) = v_C(0^+) = -30\ \mathrm{V}$

(ii) the final value of $v_C(t)$: $v(\infty) = +90\ \mathrm{V}$ (i.e., v_f)

(iii) the time constant τ : $\tau = RC = (400\mathrm{k}) \times (0.5\mu\mathrm{F}) = 0.2\ \mathrm{s}$

(iv) $v_C(t) = 90 + (-30 - 90)\, e^{-5t} = 90 - 120\, e^{-5t}\ \mathrm{V}, \quad t \geq 0$

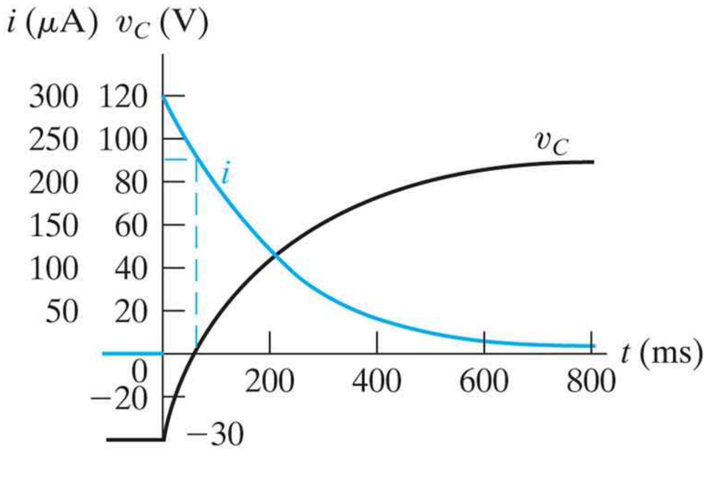
(i) the initial value of $i(t)$:

$$i(0^+) = [90 - (-30)] / (400 \times 10^3) = 300\ \mu\mathrm{A}$$

(ii) the final value of $i(t)$: $i(\infty) = 0\ \mathrm{A}$ (i.e., i_f)

(iii) the time constant is not changed : $\tau = 0.2\ s$

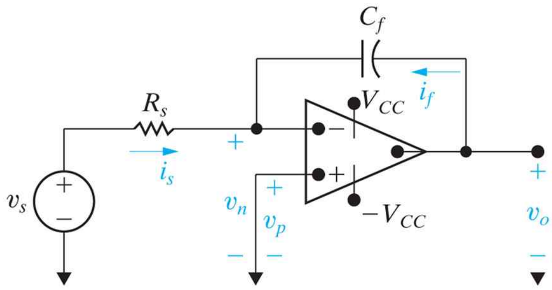
(iv) $i(t) = 0 + (300 - 0)\, e^{-5t} = 300\, e^{-5t}\ \mu\mathrm{A}, \quad t \geq 0^+$



※ the voltage across the capacitor : continuous

the current through a capacitor : not continuous

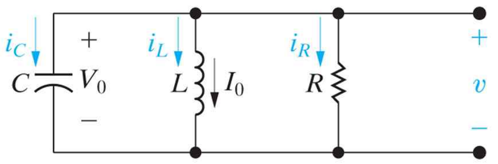
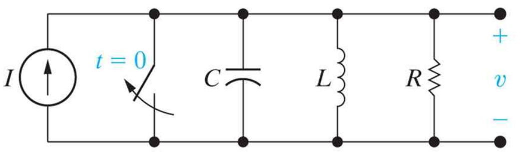
§7.7 The Integrating Amplifier

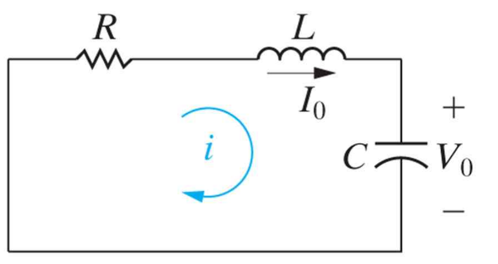
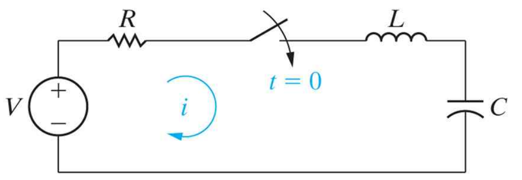


$$v_0(t) = -\frac{1}{R_s C_f}\int_{t_0}^{t} v_s(\lambda)\, d\lambda + v_0(t_0)$$

※ If we use this circuit, we can solve a first-order differential equation. That is, we can make an analog computer.

Chap.8 Natural and Step Responses of RLC Circuits

Parallel RLC circuit without sources	Parallel RLC circuit with sources
	

Series RLC circuit without sources	Series RLC circuit with sources
	

- two memory element circuits

  We must solve second-order differential equations in order to analyze circuits with two memory elements.

  - RLC circuits without sources (no inputs) :

    We must solve the second-order homogeneous diff. eqs.

    General solution only

  - RLC circuits with sources (with inputs) :

    We must solve the second-order nonhomogeneous diff. eqs.

    General solution + Particular solution

※ 2nd-order differential equations (EM textbook, Chap.4)

We begin our study of the linear second-order constant-coefficient differential equation

(1) $ay'' + by' + cy = f(t) \quad (a \neq 0)$

with the special case where the function $f(t)$ is zero:

(2) $\boldsymbol{ay'' + by' + cy = 0}$.

Because e^{rt} is never zero, we can divide by it to obtain

(3) $\boldsymbol{ar^2 + br + c = 0}$.

Consequently, $y = e^{rt}$ is a solution to (2) if and only if r satisfies equation (3). Equation (3) is called the **auxiliary equation** (also known as the **characteristic equation**) associated with the homogeneous equation (2).

Now the auxiliary equation is just a quadratic, and its roots are

$$r_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \quad \text{and} \quad r_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a} .$$

DISTINCT REAL ROOTS

If the auxiliary equation (3) has distinct real roots r_1 and r_2, then both $y_1(t) = e^{r_1 t}$ and $y_2(t) = e^{r_2 t}$ are solutions to (2) and $y(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$ is a general solution.

REPEATED ROOT

If the auxiliary equation (3) has a repeated root r, then both $y_1(t) = e^{rt}$ and $y_2(t) = te^{rt}$ are solutions to (2), and $y(t) = c_1 e^{rt} + c_2 te^{rt}$ is a general solution.

COMPLEX CONJUGATE ROOTS

If the auxiliary equation has complex conjugate roots $\alpha \pm i\beta$, then two linearly independent solutions to (2) are

$e^{\alpha t}\cos \beta t$ **and** $e^{\alpha t}\sin \beta t$,

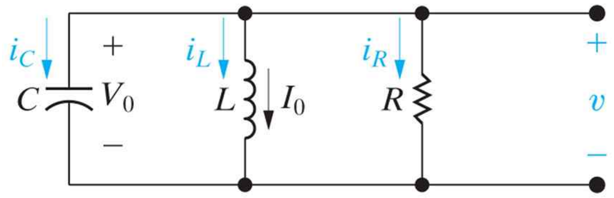
and a general solution is

(9) $\boldsymbol{y(t) = c_1 e^{\alpha t} \cos \beta t + c_2 e^{\alpha t} \sin \beta t}$,

where c_1 and c_2 are arbitrary constants.

§8.1 The Natural Response of a Parallel RLC Circuit

- the general solution of the second-order homogeneous differential equation



KCL :

$$C\frac{dv(t)}{dt}+\frac{1}{L}\int_0^t v(\lambda)\,d\lambda+I_0+\frac{v(t)}{R}=0$$

Differentiate with time and rearrange, then

$$\frac{d^2v(t)}{dt^2}+\frac{1}{RC}\frac{dv(t)}{dt}+\frac{1}{LC}v(t)=0$$

(※ $y''+ay'+by=0$; same second-order homogeneous differential equation in Engineering Mathematics course)

Let's assume that the solution is $v(t)=Ae^{st}$

$$Ae^{st}\left(s^2+\frac{s}{RC}+\frac{1}{LC}\right)=0$$

$$\therefore\quad s^2+\frac{s}{RC}+\frac{1}{LC}=0 \quad \text{(the characteristic equation)}$$

$$s_1=-\frac{1}{2RC}+\sqrt{\left(\frac{1}{2RC}\right)^2-\frac{1}{LC}}$$

$$s_2=-\frac{1}{2RC}-\sqrt{\left(\frac{1}{2RC}\right)^2-\frac{1}{LC}}$$

Therefore

$$v(t)=A_1e^{s_1t} \quad \text{and} \quad v(t)=A_2e^{s_2t}$$

The summation of two terms are also a solution.

$$v(t)=A_1e^{s_1t}+A_2e^{s_2t}$$

s_1 and s_2 can be represented as following;

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} \ , \qquad s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

where

$$\alpha = \frac{1}{2RC} \quad \text{and} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

- There are 3 cases for s_1 and s_2

(1) $\omega_0^2 < \alpha^2$ (distinct real roots : overdamped)

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

(2) $\omega_0^2 = \alpha^2$ (repeated root : critically damped)

$$v(t) = D_1 t e^{-\alpha t} + D_2 e^{-\alpha t}$$

(3) $\omega_0^2 > \alpha^2$ (complex conjugate roots : underdamped)

$$\begin{aligned} v(t) &= A_1 e^{(-\alpha + j\omega_d)t} + A_2 e^{(-\alpha + j\omega_d)t} \\ &= e^{-\alpha t} [\, (A_1 + A_2) \cos \omega_d t + j\,(A_1 - A_2) \sin \omega_d t \,] \\ &\equiv B_1 e^{-\alpha t} \cos \omega_d t + B_2 e^{-\alpha t} \sin \omega_d t \end{aligned}$$

<u>Not Easy: Study more in the Engineering Mathematics class.</u>

※ Engineering Math Textbook (§4.3 Exercise Prob.34)